LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	20
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Forces in climb, Excess power, max angle, maximum rate, Economical climb, time to climb
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Forces in climb, Excess power, max angle, maximum rate, Economical climb, time to climb.
5. Specific Objectives	6.1.3.1 Trainee will be able to describe flight performance of aircraft.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of climb performance charts; Instructor supervision required for chart interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the four primary forces acting on an aircraft in steady flight?
(b) How does the relationship between thrust and drag change during a climb?
(c) Define excess power in the context of aircraft performance.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Forces acting during climb • Excess power definition • Maximum angle of climb • Maximum rate of climb • Economical climb speed	L & Ds	PPT, IB	15 min
(b)	• Time to climb calculation • Performance chart analysis • Practical chart review • Safety precautions • Instructor supervision	L & Ds	PPT, PDF	20 min
(c)	• Summary of climb performance • Key performance parameters • Operational implications • Review of objectives • Preparation for descent	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between maximum angle of climb and maximum rate of climb.
(b) How is economical climb speed determined for fuel efficiency?
(c) Describe the role of excess power in achieving a climb.

12. Home Assignment and Follow Up	Prepare a brief note on how descent forces differ from climb forces, referencing the next lesson plan on glide range.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Descend forces in a descend glide range.	Refer to the master syllabus reference list	LP 21

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min